(DEEMED TO BE UNIVERSITY)

INTELLIGENT EMAIL MANAGEMENT AND SPAM DETECTION PLATFORM

Database Engineering & Database Systems Design (DBE & DSD) | Project Review – 1

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PROBLEM STATEMENT



Spam emails clutter inboxes, making it difficult to identify genuine emails. They increase the risk of phishing, fraud, and malicious attacks posing serious threats to individuals, businesses, educational institutions, and organizations worldwide.



Efficient storage and management of email data is essential. Without a structured database system, tracking, classifying, and retrieving emails becomes slow, error-prone, and insecure highlighting the urgent need for a robust DBMS-based solution.

NEED FOR THE PROJECT

Email Users
Efficient email management and organized storage of email records for individuals and businesses.
Reduces manual effort in sorting spam from genuine emails.

Companies
Secure management of spam information and faster email retrieval for corporate environments.
Improved database management for enterprise-level email systems.

Educational Institutions
Organized email data storage and efficient retrieval for academic organizations.
Better classification of communications across departments.

PROJECT OBJECTIVES



To design a relational database for efficient email management and store user and email information in a structured, normalized schema for optimal performance and data integrity.



To maintain comprehensive spam detection records and support quick email search and retrieval using indexed queries and optimized relational database structures.



To demonstrate core SQL operations including joins, triggers, and stored procedures while applying database normalization principles to ensure a scalable and efficient system.

PROPOSED METHODOLOGY

User Registration / Login

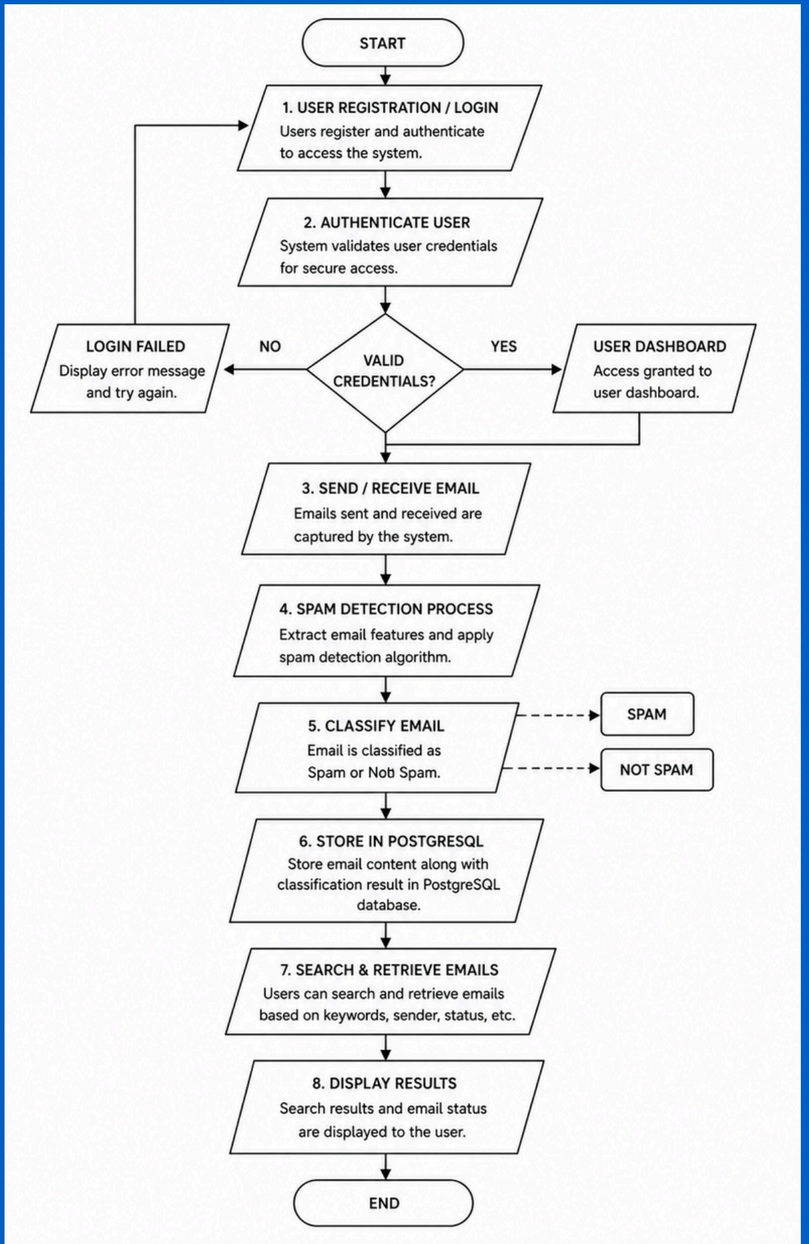
Users register and authenticate securely. Credentials and profile data are stored in the PostgreSQL database for session management and access control.

Send / Receive Email & Store in PostgreSQL

Emails sent and received are captured and stored in a structured relational database. Spam Detection results are recorded alongside each email entry.

Search & Retrieve Emails

Modules: User Management | Email Management | Spam Detection Management | Search & Classification | Database Testing & Documentation



FEASIBILITY ANALYSIS

Hardware Requirements

- Laptop
- Windows 11 OS
- intel core i7
- 16 GB RAM
- Sufficient storage for PostgreSQL database and project files
- Standard internet connectivity for testing and collaboration

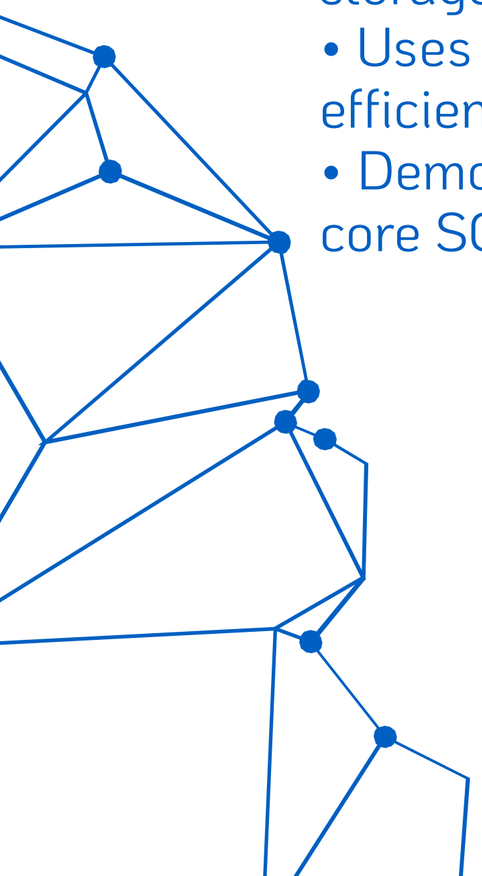
Software & Tools

- PostgreSQL
- VS Code (Editor)- Python Version -3.14 Java(DSA)
- Web Browser for interface testing and project documentation

Language & Dataset

- Programming Language: SQL
- Self-created sample email dataset
- Normalized relational schema with realistic spam and genuine email records

NOVELTY & EXPECTED OUTCOME



Novelty

- Combines email management with spam record storage in one unified system
- Uses a normalized relational database for efficient, non-redundant data
- Demonstrates real-world DBMS concepts using core SQL operations

Expected Outcome

- A secure, efficient email management and spam detection system
- Demonstrates database design, normalization & SQL operations
- Enables fast, reliable retrieval of email and spam records

THANK YOU